

Smart Health & Fitness Tracker — API Endpoints Summary

Module: COM661 Full Stack Strategies and Development **Assignment:** CW2 (Front-End)

Back-end base URL: `http://127.0.0.1:5001/api` **Authentication:** JWT Bearer token (24 h expiry)

Note on back-end modifications: the CW1 Flask/MongoDB API is used **unchanged**. No endpoints have been added, removed or modified for CW2. Only the front-end Angular application developed for this coursework will contribute to the mark.

1. Authentication

Method	Endpoint	Auth	Angular caller	Purpose
POST	/register	No	AuthService.register()	Create a new account (email, password, profile fields). Auto-logs user in and returns JWT.
POST	/login	No	AuthService.login()	Authenticate and return a JWT. Consumed by LoginComponent; token persisted in localStorage.

2. User Profile

Method	Endpoint	Auth	Angular caller	Purpose
GET	/profile	Yes	UserService.getProfile()	Fetch signed-in user's profile. Used by ProfileComponent on load.
PUT	/profile	Yes	UserService.updateProfile()	Partial update of name, age, height, weight, gender, membership.
DELETE	/profile	Yes	UserService.deleteProfile()	Permanently delete own account (followed by auto-logout).

3. Activity Logs

Method	Endpoint	Auth	Angular caller	Purpose
POST	/activities	Yes	ActivityService.create()	Log a new workout. Called from ActivityFormComponent.
GET	/activities	Yes	ActivityService.list()	Paginated list with filters (activity_type, start_date, end_date, page, per_page). Query-string params sent via HttpParams.
GET	/activities/{id}	Yes	ActivityService.get()	Retrieve a single entry (pre-population for edit form).
PUT	/activities/{id}	Yes	ActivityService.update()	Edit an existing activity.
DELETE	/activities/{id}	Yes	ActivityService.delete()	Remove entry (triggers confirmation dialog in list view).

4. Nutrition Logs

Method	Endpoint	Auth	Angular caller	Purpose
POST	/nutrition	Yes	NutritionService.create()	Log a new meal with optional macros.
GET	/nutrition	Yes	NutritionService.list()	Paginated list filterable by meal_type + date range.

GET	/nutrition/{id}	Yes	NutritionService.get()	Single entry for edit pre-population.
PUT	/nutrition/{id}	Yes	NutritionService.update()	Edit macros/calories/food name.
DELETE	/nutrition/{id}	Yes	NutritionService.delete()	Delete entry.

5. Health Metrics

Method	Endpoint	Auth	Angular caller	Purpose
GET	/health-metrics	Yes	HealthMetricsService.get()	Current vitals (HR, BP, sleep, VO2) plus server-computed BMI.
PUT	/health-metrics	Yes	HealthMetricsService.update()	Partial update; null fields preserved.

6. Fitness Goals

Method	Endpoint	Auth	Angular caller	Purpose
GET	/goals	Yes	GoalsService.get()	Fetch current fitness targets.
PUT	/goals	Yes	GoalsService.update()	Set/update target weight, calorie target, workouts/week, goal type, target date.
DELETE	/goals	Yes	GoalsService.clear()	Clear all goals (with confirmation).

7. Analytics

Method	Endpoint	Auth	Angular caller	Purpose
GET	/analytics/bmi-report	Yes	AnalyticsService.bmiReport()	BMI with category + advice. Shown on Dashboard + Analytics page.
GET	/analytics/calorie-balance	Yes	AnalyticsService.calorieBalance()	Intake vs burned for date range (querystring start_date / end_date).
GET	/analytics/weekly-progress	Yes	AnalyticsService.weeklyProgress()	Day-by-day breakdown for charts (bar chart on Dashboard, line on Analytics).
GET	/analytics/goal-tracking	Yes	AnalyticsService.goalTracking()	Progress vs targets. Powers progress bars on Dashboard.
GET	/analytics/activity-summary	Yes	AnalyticsService.activitySummary()	MongoDB aggregation by activity type. Doughnut chart on Analytics.
GET	/analytics/nutrition-summary	Yes	AnalyticsService.nutritionSummary()	MongoDB aggregation by meal type. Stacked-bar chart on Analytics.
GET	/analytics/admin/top-active-users	Admin	AnalyticsService.topActiveUsers()	Leaderboard (querystring limit). Horizontal-bar chart on Admin page.

8. Admin — User Management

Method	Endpoint	Auth	Angular caller	Purpose
GET	/users	Admin	UserService.listUsers()	Paginated + filterable (search, membership_type, gender).
GET	/users/{id}	Admin	UserService.getUser()	Get any user by ID.
DELETE	/users/{id}	Admin	UserService.deleteUser()	Delete any user (self-delete blocked client-side).
PUT	/users/{id}/role	Admin	UserService.changeRole()	Toggle user/admin role with confirmation.

HTTP methods in use (rubric evidence)

Method	Count	Example caller
GET	18	ActivityService.list(), AnalyticsService.bmiReport(), ...
POST	3	AuthService.login(), register(); ActivityService.create()
PUT	7	UserService.updateProfile(), GoalsService.update(), ...
DELETE	5	ActivityService.delete(), GoalsService.clear(), UserService.deleteUser(), ...

All four HTTP verbs are actively used, satisfying the **60-69% Communication with Back-end** criterion. Querystring parameters (pagination, date ranges, filters) and request-header parameters (JWT Bearer) satisfy the **70-79% flexible requests** criterion. Combined client/server JWT handling — `HttpInterceptor` + functional guards + server-issued token — satisfies the **80%+ integrated authentication** criterion.

Querystring parameters summary

Endpoint	Parameter	Type	Purpose
/activities	activity_type	string	Filter by workout type
/activities	start_date / end_date	YYYY-MM-DD	Date range
/activities	page / per_page	integer	Pagination
/nutrition	meal_type	string	Filter by meal
/nutrition	start_date / end_date	YYYY-MM-DD	Date range
/nutrition	page / per_page	integer	Pagination
/users	search	string	Admin: fuzzy full_name
/users	membership_type / gender	string	Admin: filters
/users	page / per_page	integer	Admin: pagination
/analytics/calorie-balance	start_date / end_date	YYYY-MM-DD	Range for balance
/analytics/admin/top-active-users	limit	integer	Leaderboard size